

Stat 5810/6810 — Project 1 (50 Points) — Fr 11/02/2012

General Instructions

Due: Th 11/15/2012, 11:59pm (by e-mail)

For this project, you will work with real data files that are recorded in the context of near-infrared spectroscopy (NIRS). NIRS is an optical imaging method for studying brain activity. In a NIRS-based experiment, an array of fiber-optic probes is placed on the head of a human participant (see Figure 1 for details).

While the cap that is worn is fixed for each human participant in a particular study, the exact dimensions of the head may differ considerably among participants. Therefore, head measurements for each participant must be obtained first. These measurements are stored in a so-called `.pos` file.

At this time, there exists no R package that allows us to read in data from these `.pos` files for further processing in R. It is our task to write functions that read in the data from the `.pos` files and provide some basic visualization of these data.

You are allowed to work in groups of up to two students. A final report in $\text{\LaTeX}$ with your figures, a summary of your results, main features of your R code you would like to highlight, and the R code in the appendix must be turned in as a `.pdf` file. I also need the `.R` file!

Below are the specific instructions:

1. The `.pos` files are plain ASCII file that can be opened in any text editor. You should make yourselves familiar with the different files that are being provided. What is identical, what is different in those files that are provided?
2. Write a function that reads in a `.pos` file and creates a list with the content of this file. You should use regular expressions wherever possible to extract the content from the `.pos` file. Your function should be called in the following way:

```
Participant1 = NIRSReadPosFile(filename = "0320NLA18F.pos", processed = TRUE)
```

`filename` will be a required parameter. Your function should stop and report an error if a file with this name does not exist.

`processed` is an optional parameter and it can take values `TRUE` and `FALSE`. The default is `TRUE`. Everything else (other than `TRUE` or `FALSE`) should result in a warning, but this should be interpreted as `TRUE` and the execution of the code should be continued. Thus, only `FALSE` is really interpreted as `FALSE`. The effect of this parameter will be described in more detail later on.

Your main list with all the data should be structured as a combination of several sub-lists and matrices. The first sub-list should be called `processed` and simply contain



Figure 1: Photos of infants who were in our experiments. The photos on the top show the cap with chin strap and elastic bandage combination that provides a snug fit between the probes and the infant's head. The pictures on the bottom represent the placement of the probes over the auditory and visual cortices (on the left and right respectively). The red circles indicate the approximate placement of the source probes and the blue circles represent the detector probes. The black squares indicate where the NIRS system would detect blood flow between each source-detector pairing. This particular probe arrangement consists of two sets of 3×3 optode arrays comprising 24 channels that each detect changes in oxy- and deoxyhemoglobin. [Figure and caption taken from p. 7 of Hespos, S. J. (2010) "What is Optical Imaging?", *Journal of Cognition and Development* 11(1):3-15.]

TRUE or FALSE, according to the setting of this parameter. The second sub-list should be called `FileInfo` (with elements such as `ID`, `VER`, etc.). The third sub-list should be called `User` (with elements such as `Name`, `ID`, etc.). The next few sub-lists should be called `Probe1`, ..., `ProbeN`, and `MRI`.

This should be followed by a matrix called **Markers**. The rows should be named **LeftEar**, **RightEar**, **Nasion**, **Back**, and **Top**. The columns should be named **X**, **Y**, and **Z**. This structure is fixed.

This should be followed by another matrix called **Probes**. The rows should be named **Probe i ch j** . The columns should be named **X** to **NZ**. This structure is variable with respect to the number of rows.

Finally, there will be two more sub-lists, called **Angle** and **Measure**.

You cannot assume that the file format always is exactly the same as in your sample files. Obviously, the number of probes and channels can change. Moreover, the manufacturer of the device may decide to write additional information into this file. Surprisingly, the serial number of the device and date and time of the recordings are not stored in this file. Thus, it could easily happen that a future version of this file contains entries such as **SerialNumber**, **Date**, and **Time**. Therefore, you cannot assume that each of the main components described above always appears in exactly the same line of the **.pos** file. Rather, you have to assume that **[User]** may occur in line 8 in one data file and in line 10 in the next **.pos** file. Thus, your code has to be flexible.

As we do not know how future additional components will be named, we have to obtain the names for the components of our sub-lists from the data. Then, if there will be a future entry called **SerialNo** (and not **SerialNumber**), this will nevertheless get processed properly. The following code snippet shows how you can add names to a list after it has been created:

```
> xx = list(c(1:4), c("a", "b"))
> names(xx) = c("Numbers", "Letters")
> xx$Numbers
[1] 1 2 3 4
> xx$Letters
[1] "a" "b"
```

3. Let us now describe the **processed** parameter in more detail. If it is **FALSE**, then all data will be saved exactly as stored in the **.pos** file. This means that a missing entry after the “=” will be stored as an empty string, the string “ 18y” for Age will be saved as is, and all numbers will also be saved as strings. This format will allow us to check whether we are reading the data correctly, in particular that we do not mix up any lines and identifiers.

Otherwise, if **processed** is **TRUE** (or anything different from **FALSE**), we will further process, i.e., transform, the data according to these rules:

- No entry after the “=” or an arbitrary number of spaces will be interpreted as missing (**NA**).

- Space at the start or end should be suppressed, i.e., “ Abc ” should be saved as “Abc”.
- For the **Sex** field, entries **Female** and **Male** should be translated to **F** and **M**, respectively. This should happen, based on the first letter, i.e., **Femal** still should be translated to **F**. The same should be done for lower case versions of the genders. If this field is empty, the **NA**-rule from above applies. If this field contains anything else that cannot be transformed into **F** or **M**, a warning should be issued and **NA** should be assigned.
- For the **Age** field, an entry such as “ 18y” simply should be translated into the number 18.
- All apparent numbers should be translated into numeric. This includes data for the probes, the angles, and several variables early in the file. Note that some unusual data entries exist, e.g., “-0.0”, which apparently should be 0.

Extensively test your **NIRSReadPosFile** function. Also create copies of the data **.pos** files and manipulate those manually, e.g., by adding or deleting some variables. You should also modify some of the content to check whether your error-checking works properly.

4. Let’s not forget that these data originate from a human head. We want to display the data in a 3-dimensional (3D) plot to check that the measurements are plausible. Install the R package *rgl* and read its documentation, in particular for the 3D plot command. When we issue a command such as

```
NIRSPlot3D(Participant1)
```

the following should happen:

- If the **processed** list is **FALSE**, we stop with an error message. Nothing will be plotted.
- Otherwise, all 3D **X**, **Y**, **Z** coordinates from **Probeichj** should be plotted in gray. Also, the 3D **X**, **Y**, **Z** coordinates from the five markers should be plotted as well with slightly bigger symbols. Select the colors from a qualitative color scheme from **RColorBrewer** with three different colors. Use one color for **Nasion** and **Back**, another color for the ears, and the third color for **Top**.

For each of the provided **.pos** files, produce four snapshots for your report showing different 3D rotations. Do not mix up the files and snapshots. It would be best to label them similar to the corresponding data file, e.g., **0320NLA18F_snap1.jpg**. Your figure caption also should specify to which data file these snapshots belong.

Good luck!!!